

AI & Machine Learning Beginner Notes

1. Artificial Intelligence (AI)

- **Definition:** Simulation of human intelligence in machines.
- **Examples:** Siri, Alexa, Netflix recommendations, self-driving cars, ChatGPT.
- **Types:**
 - **Narrow AI:** Specialized in one task (e.g., spam filter, recommendation systems).
 - **General AI:** Can perform any intellectual task (not yet achieved).

2. Machine Learning (ML)

- **Definition:** Subset of AI where machines learn from data without being explicitly programmed.
- **Why ML Matters:** Predicts outcomes, automates tasks, extracts insights from data.

Types of ML

1. **Supervised Learning:** Learn from labeled data.
 - Regression (predict numbers): Linear Regression
 - Classification (predict categories): Logistic Regression, Decision Trees
2. **Unsupervised Learning:** Learn patterns without labels.
 - Clustering: K-Means
 - Dimensionality Reduction: PCA
3. **Reinforcement Learning:** Learn through trial and error using rewards.
 - Example: Self-driving cars, game-playing AI

3. Key ML Algorithms

Algorithm	Type	Example
Linear Regression	Regression	House price prediction
Logistic Regression	Classification	Spam detection
Decision Trees	Both	Loan approval
Random Forest	Both	Credit risk models
SVM	Classification	Image classification
KNN	Both	Recommender systems
K-Means	Unsupervised	Customer segmentation
PCA	Unsupervised	Dimensionality reduction
Naive Bayes	Classification	Sentiment analysis
Neural Networks	Deep Learning	Image recognition, ChatGPT

4. How a Machine Learns

1. Give the model data (features + labels for supervised).
2. Model makes predictions.
3. Compute error (loss function).
4. Adjust parameters (Gradient Descent) to reduce error.

5. Repeat until predictions are accurate.

5. Python Basics for ML

- **Variables & Data Types:** int, float, str, bool, list
- **Lists & Loops:** Store and iterate data.
- **Functions:** Reusable code blocks.
- **If-Else:** Conditional logic.
- **Libraries:**
 - **NumPy:** Numerical operations and arrays.
 - **Pandas:** Load and manipulate datasets.
 - **Matplotlib:** Data visualization.

6. Supervised vs Unsupervised Learning

Feature	Supervised	Unsupervised
Labels	Yes	No
Goal	Predict output	Find patterns
Examples	House price, spam detection	Customer segmentation, anomaly detection

7. Linear Regression

- Predicts numerical outcomes.
- Equation: $y = m \cdot x + b$
- Learns **slope (m)** and **intercept (b)** to minimize error.

8. Key Takeaways

- Start with **Python basics + NumPy + Pandas + Matplotlib**.
- Learn **Supervised Learning algorithms** first: Linear Regression, Logistic Regression, Decision Trees, Random Forest.
- Understand **how machines learn** through prediction, error calculation, and parameter adjustment.
- Explore **unsupervised algorithms** (K-Means, PCA) later.
- Neural Networks and Deep Learning come after mastering basics.

****End of Notes****